

CARLOS PEREZ MENDOZA

Cell phone. [+1 438 346 2908](tel:+14383462908) | carlos.pmendoza3@gmail.com
<https://cpmendoza.github.io> | www.linkedin.com/in/octaviopm

Professional summary

Research Scientist with a Ph.D. in Mathematics & Statistics and 8+ years of experience in quantitative finance and actuarial science. I develop and apply machine learning, reinforcement learning, and probabilistic modeling to create new methods for risk management, financial derivatives, and insurance, supported by a strong record of publications and international collaborations.

Education

Doctor of Philosophy in Mathematics and Statistics GPA: 4.3/4.3 Concordia University, Department of Mathematics & Statistics	Canada Sep 2021-Apr 2025
Master of Science in Modern Applications of Mathematics / Visiting researcher student. University of Bath, Department of Mathematical Sciences	United Kingdom Jan 2020-Jul 2020
Master of Science in Mathematics, Mathematical Statistics and Probability GPA: 9.9/10 Institute of Applied Mathematics and Systems Research, UNAM	Mexico Jan 2019-Jul 2020
Specialist Degree in Financial Engineering , GPA: 9.6/10 Universidad Nacional Autónoma de México, UNAM - Faculty of Engineering.	Mexico Aug 2021-Jul 2022
BSc. in Actuarial Sciences, Actuarial and Financial Mathematics , GPA: 9.20/10 Universidad Nacional Autónoma de México, UNAM - Faculty of sciences	Mexico Aug 2010-Dec 2014
Graduate Diploma in Statistical Techniques and Data Mining , GPA: 10/10 Universidad Nacional Autónoma de México, UNAM	Mexico Jan 2016-Oct 2016

Academic Positions

Postdoctoral Fellow McGill University	Montreal, Canada Since Jul 2025
Research Fellow , Department of Management Sciences HEC Montréal (Université de Montréal's Business School)	Montreal, Canada May 2025-Aug 2025
Instructor of Machine Learning Courses , Continuing Education for Professionals Université de Montréal	Montreal, Canada Jan 2024-Jan 2025
Instructor of Undergraduate Mathematics Courses , Department of Mathematics and Statistics Concordia University	Montreal, Canada Aug 2021-Mar 2025
Lecturer , BSc. in Actuarial Sciences Universidad Autónoma del Estado de México, UAEM	Mexico City, Mexico Jan 2019-Aug 2021
Lecturer , BSc. in Actuarial Sciences Universidad Nacional Autónoma de México, UNAM	Mexico City, Mexico Aug 2015-Aug 2021
Postgraduate teaching assistant , M.Sc. in Mathematics & Spec. in Applied Statistics Research Institute in Applied Mathematics and Systems, IIMAS - UNAM	Mexico City, Mexico Oct 2020-Aug 2021

Professional Experience (Industry)

JACOB - Centre d'intelligence artificielle appliquée, Direction Recherche & Innovation Scientifique de recherche appliquée	Montreal, Canada Mar 2023 – Jul 2024
Plant-E Corp, Algorithmic Trading Department Quantitative Research Intern	Montreal, Canada Jul 2022-Nov 2022

Private Assistance Board of the Government of Mexico City

Strategic Information Management Coordinator

Mexico City

Mar 2021-Ago 2021

Private Assistance Board of the Government of Mexico City

Presidential Advisor

Mexico City

Oct 2020-Mar 2021

Bank of Mexico (Central Bank of Mexico), Technical Sub-Management of the Health Plan

Head of the Office of Quantitative Health Plan Studies

Mexico City

Nov 2017-Jan 2019

Bank of Mexico (Central Bank of Mexico), Health Plan Evaluation Sub-Management

Market Research Analyst

Mexico City

Mar 2015-Oct 2017

Research reports submitted for publication in peer-reviewed journals

1. **Pérez-Mendoza, C. O.** , G., Godin, F. (2025). Hedging Targeted Risks with Reinforcement Learning: Application to Life Insurance Contracts with Embedded Guarantees, submitted to *ASTIN Bulletin: The Journal of the IAA*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5279418
2. François, P., Gauthier, G., Godin, F., **Pérez-Mendoza, C. O.** (2025). Deep Hedging with Options using the Implied Volatility Surface, submitted to *Management Science*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5209935
3. François, P., Gauthier, G., Godin, F., **Pérez-Mendoza, C. O.** (2024). Is the difference between deep hedging and delta hedging a statistical arbitrage?, *Finance Research Letters*, [73:106590](https://doi.org/10.1016/j.frl.2024.106590).
4. François, P., Gauthier, G., Godin, F., **Pérez-Mendoza, C. O.** (2024). Enhancing Deep Hedging of Options with Implied Volatility Surface Feedback Information, submitted to *Mathematical Finance*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4910867

Working papers

1. **Pérez-Mendoza, C. O.**, Siri-Jégousse, A., Gutiérrez-Peña, E., Riva-Palacio, A. (2025). Parameter estimation for partially observed stable continuous-state branching processes.

Conferences

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| <i>Hedging Targeted Risks with Reinforcement Learning: Application to Life Insurance Contracts with Embedded Guarantees</i> , 60 th Actuarial Research Conference. | Toronto
Jul 2025 |
| <i>Enhancing Deep Hedging of Options with Implied Volatility Surface Feedback Information</i> , Minnesota Center for Financial and Actuarial Mathematics, University of Minnesota. | Minnesota, USA
Sep 2024 |
| <i>Financial simulation of the cost of health services at retirement age</i> , National Autonomous University of Mexico, Faculty of Higher Studies Acatlán, 3rd Financial Colloquium. | Mexico City
Apr 2018 |
| <i>Binary logistical models such as epidemiological profiling technique</i> , National Autonomous University of Mexico, Faculty of Higher Studies Acatlán, 6th Statistical Colloquium. | Mexico City
Feb 2018 |
| <i>Financial projection in the health sector via stochastic simulation</i> , Autonomous University of the State of Mexico, 3rd International Congress of Actuarial Science and Economics. | Mexico City
Nov 2017 |
| <i>Markov processes in the evaluation of health prevention projects</i> , National Autonomous University of Mexico, Faculty of Higher Studies Acatlán, 5th Statistical Colloquium. | Mexico City
Mar 2017 |
| <i>Population classification and risk segmentation in the health sector</i> , National Autonomous University of Mexico, Faculty of Higher Studies Acatlán, XVII Academic Week of the career of Actuary. | Mexico City
Sep 2016 |
| <i>Poisson-Gamma Model as identifier of atypical behaviors</i> , National Autonomous University of Mexico, Faculty of Higher Studies Acatlán, 4th Colloquium of Statistics. | Mexico City
Mar 2016 |

Recognitions

- **MITACS Elevate Postdoctoral Fellowship Program** – Prestigious postdoctoral fellowship supporting collaborative research and leadership development. Jul 2025-Jul 2026
- **CIA Academic Research Grant** – Awarded by the Canadian Institute of Actuaries to support PhD research. Apr 2024-Apr 2025
- **MITACS Accelerate Program** – Industry-academic internship funding for PhD students. Jun 2022-Nov 2022
- **FIN-ML Graduate Fellowship (NSERC-CREATE Grant at the Doctorate Level)** – Competitive fellowship for AI/ML research in financial applications. Sep 2021-Sep 2022
- **CONACYT National Postgraduate Fellowship** – Awarded by the Mexican government for top graduate students. Feb 2019-Feb 2021
- **Top 3 GPA Award** – Awarded for ranking among the top three students in Actuarial Science, *Universidad Nacional Autónoma de México*. Aug 2010-Dec 2014
- **Student of Excellence Award** - Conferred by the Government of the State of Mexico for outstanding academic performance. Jul 2012

Workshops & Volunteer Experience

Université de Montreal & CIMAT	Canada - Mexico
Workshop - Machine Learning applied to Industry	Jan 2023 - Feb 2023
Université de Montreal, UdeM	Canada
Workshop - Machine Learning applied to Finance and Insurance	Nov 2021 - Dec 2021
Massachusetts Institute of Technology, MIT	United States of America - India
Volunteer for Mentoring	Jul 2020, Sep 2020
University of Bath	United Kingdom - Mexico
Workshop - Integrative Think Tank	Jul 2019 - Ago 2019
National Autonomous University of Mexico, UNAM	Mexico City
Volunteer for Mentoring	Jan 2020 - Sep 2020